ELEC70091
EE9-CS2-1

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

MSc and EEE/EIE PART IV: MEng and ACGI

STABILITY AND CONTROL OF NON-LINEAR SYSTEMS

Wednesday, 15 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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1. Consider the following system of first order nonlinear differential equations:

$$\begin{aligned} \dot{x}_1 &= (6-x_1)(6+x_1)x_1 \\ \dot{x}_2 &= -x_2^3+x_2+x_1. \end{aligned}$$

Let $x=[x_1,x_2]'$ denote the state of the system, evolving in $\mathbb{R}^2$.

a) Discuss existence and uniqueness of solutions. [2]

b) Find analytically and sketch graphically the nullclines of the vector-field (Hint: for nullcline 2 determine the local minima and maxima of the nullcline when interpreted as a function of x_2; this allows for a better sketch). [2]

c) Show that nullcline $\mathcal{N}_1$ (corresponding to $\dot{x}_1=0$) is an invariant set. [2]

d) Find regions in $\mathbb{R}^2$ delimited by nullclines and identify the orientation of the vector-field in each one of them. [2]

e) Find all forward invariant regions among those identified in part d). [2]

f) Compute all equilibria of the system, linearize the system around each equilibrium and discuss the local phase-portrait. [4]

g) Sketch a global phase-portrait for the system. [3]

h) Discuss the ω-limit set $\omega(x_0)$ for $x_0 \neq 0$. [3]

2. Consider the following second-order differential equation:

$$\ddot{z}(t) = -[2 + \sin(z(t))]z(t) - \dot{z}(t) + u(t),$$

where $z(t)$ and $u(t) \in \mathbb{R}$.

a) Find a state-space representation of the system, with input $u(t)$ and output $y(t) := \dot{z}(t)$. [3]

b) Highlight that the system considered is input-affine. [2]

c) Show, using the passivity criterion for input-affine systems, that a storage function to prove passivity from u to y can only be of the form:

$$\frac{\dot{z}^2}{2} + P(z)$$

for some suitable differentiable function $P : \mathbb{R} \to \mathbb{R}$. [3]

d) Find, using the state-space representation introduced in question a), a storage function $S(x)$ to prove the system is passive from u to y. [3]

e) Consider the feedback $u = 2y$. Assess the stability of the origin for the resulting closed-loop system. [3]

f) Consider next the feedback $u = -2y$. Prove that the resuting closed-loop system is Globally Asymptotically Stable using a suitable candidate Lyapunov function and stability criterion. [6]

3. Consider the following system of differential equations:

$$\begin{aligned} \dot{x}_1 &= -3x_1 + \text{atan}(x_2), \\ \dot{x}_2 &= \text{atan}(-x_2 + 2x_1). \end{aligned}$$

a) Interpret the system as a feedback interconnection of two nonlinear scalar systems Σ_1 and Σ_2. Write the "open-loop" equations of each system with corresponding input signals and explicitly write down the equations that define the interconnection. [4]

b) Show that system Σ_1 is Input-to-State Stable (ISS) with respect to its scalar input, and compute the corresponding ISS gain (Hint: try to be as tight as possible in deriving a suitable gain function). [4]

c) Show that system Σ_2 is Input-to-State Stable with respect to its scalar input, and compute the corresponding ISS gain. [4]

d) Exploit the previous analysis to argue that the system is globally asymptotically stable (GAS) at the origin. [4]

e) Would the conclusion in question d) still be true replacing $-x_2$ with $+x_2$ in the equation for $\dot{x}_2$? Justify your answer. [4]

4. Consider the SISO nonlinear control system of equations:

$$\begin{aligned} \dot{x}_1 &= x_2 + \text{atan}(x_3) \\ \dot{x}_2 &= -2x_2 + x_1^3 \\ \dot{x}_3 &= \cos(x_2) + u \\ \\ y &= x_1, \end{aligned}$$

with input u, output y and state $x = [x_1, x_2, x_3]' \in \mathbb{R}^3$.

a) Compute, if possible, the relative degree at $x = 0$, and state whether it is locally or globally defined. [3]

b) Design an input-output linearizing state-feedback law $u(t) = k(x(t), v(t))$, where $v(t)$ is the new auxiliary scalar input and k is a suitably chosen function. [3]

c) Bring the system in normal form, and highlight the internal dynamics, by choosing a suitable state variable to complement the output and its derivative in the definition of the desired change of variables. [6]

d) The internal dynamics are forced by y, which can be regarded as an exogenous input. Discuss the ISS of the internal dynamics with respect to the input signal y. [3]

e) Design a linear feedback stabilizer for the equilibrium corresponding to the output $y = 0$ and comment on its global or local stability properties. [2]

f) Would the conclusion of the analysis carried out so far be modified if the equation for $\dot{x}_2$ is modified as follows: $\dot{x}_2 = 2x_2 + x_1^3$? (justify your answer) [3]